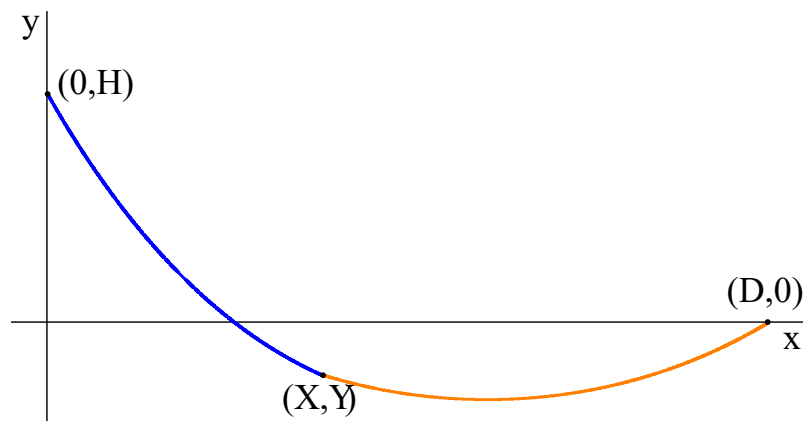


< Previous	🎥 ✓	🎥 ✓	📝 ✓	📝	📄	🎥	Next >
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4. Add a stationary rider

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Now consider the case where we do have a rider on the zipline. However, we are not ready to consider the dynamic case, and instead reduce to a static situation. We apply a force to the zipline rider so that the rider is fixed in place. (The cable may still move freely through the zipline pulley. You can imagine this by actually holding the rider in place, or using a string or some other mechanism to apply a force to the rider.)



There are six conditions to satisfy.

1. The left catenary must pass through the point (0,H).
2. The left catenary must pass through the rider position (X,Y).
3. The right catenary must pass through the point (D,0).
4. The right catenary must pass through the point (X,Y).
5. The arc length must be constant.
6. The tension $|T|$ must be continuous at the rider (torque balance).

(Note that this last condition is true because we are assuming a simple model where we have ignored effects due to friction in the pulley. If we allow this model to have friction in the pulley, then tension is no longer continuous at the rider, but has a jump discontinuity proportional to the friction term.)

Below we demonstrate MATLAB code to solve this system. Fill in the equations that determine the first four conditions.

Solving boundary conditions with a stationary rider (External resource)

(1.0 points possible)

$$y_L = c(3) + c(2) \cosh\left(\frac{x - c(4)}{c(2)}\right)$$
$$y_R = c(6) + c(5) \cosh\left(\frac{x - c(4)}{c(5)}\right)$$

All you are required to do is define the first four boundary conditions in terms of these constants and the given positions of the left endpoint

Script ?

Save

Reset

MATLAB Documentation (https://www.mathworks.com/help/)

```
1 %Read image
2 I = imread('up.jpg');
3
4 %Endpoints
5 L = [0.1464,2.533]; %(x,y) coordinate of left endpoint of cable
6 R = [5.366,1.522]; %(x,y) coordinate of right endpoint of cable
7 M = [2.509,1.669]; %(x,y) coordinate of rider
8 len = 6.05; %length of cable
9
10 %Show image and indicate endpoints and rider
11 imagesc([0,5.5098],[0,3.0993],flipud(I));set(gca,'YDir','normal');
12 axis equal; ylim([0,3.0993]); xlim([0,5.5098]); box on; hold on;
13 xlabel('$x$ (meters)','interpreter','latex'); ylabel('$y$ (meters)','interpreter','latex');
14 plot(L(1),L(2),'o','MarkerEdgeColor','b','MarkerFaceColor','b','MarkerSize',8);
15 plot(R(1),R(2),'o','MarkerEdgeColor','b','MarkerFaceColor','b','MarkerSize',8);
16 plot(M(1),M(2),'o','MarkerEdgeColor','b','MarkerFaceColor','b','MarkerSize',8);
17
18 %Define system of 6 equations to solve (c is vector containing 6 unknowns)
19 F1 = @(c) c(3) + c(2)*cosh((L(1)-c(1))/c(2)) - L(2); %Left cable must pass through left endpoint L
```

```
20 F2 = @(c) c(3) + c(2)*cosh((M(1)-c(1))/c(2)) - M(2); %Left cable must pass through rider at M
21 F3 = @(c) c(6) + c(5)*cosh((M(1)-c(4))/c(5)) - M(2); %Right cable must pass through rider at M
22 F4 = @(c) c(6) + c(5)*cosh((R(1)-c(4))/c(5)) - R(2); %Right cable must pass through right endpoint R
23 F5 = @(c) c(2)*(sinh((M(1)-c(1))/c(2))-sinh((L(1)-c(1))/c(2))) + c(5)*(sinh((R(1)-c(4))/c(5))-sinh((M(1)-
24 F6 = @(c) c(2)*cosh((M(1)-c(1))/c(2))- c(5)*cosh((M(1)-c(4))/c(5)); %Tension must be continuous at rider
25 cOut = fsolve(@(c) [F1(c);F2(c);F3(c);F4(c);F5(c);F6(c)],[-1,1,1,-1,1,1]);
26
27 %Plot theoretical shape of cable
28
29 %Left cable
30 xPlot = linspace(L(1),M(1),25);
31 yPlot = cOut(3) + cOut(2)*cosh((xPlot-cOut(1))/cOut(2));
32 plot(xPlot,yPlot,'r:','linewidth',4,'markersize',10);
33
34 %Right cable
35 xPlot = linspace(M(1),R(1),25);
36 yPlot = cOut(6) + cOut(5)*cosh((xPlot-cOut(4))/cOut(5));
37 plot(xPlot,yPlot,'b:','linewidth',4,'markersize',10);
```

[▶ Run Script](#)

Assessment: All Tests Passed

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✔ cOut correct

Output

Equation solved (matlab: helpview('optim','eqn_solved','CSHelpWindow'))).

fsolve completed because the vector of function values is near zero
as measured by the value of the function tolerance (matlab: helpview('optim','fcn_tolerance_fsolve',
the problem appears regular (matlab: helpview('optim','appears_regular','CSHelpWindow')) as measured

A stationary rider, more realistic position (External resource) (1.0 points possible)

Script

[Save](#)[Reset](#)[MATLAB Documentation \(https://www.mathworks.com/help/\)](https://www.mathworks.com/help/)

```
1 %Read image
2 I = imread('down.jpg');
3
4 %Endpoints
5 L = [0.1464,2.533]; %(x,y) coordinate of left endpoint of cable
6 R = [5.366,1.522]; %(x,y) coordinate of right endpoint of cable
7 M = [2.492,0.6204]; %(x,y) coordinate of rider
8 len = 6.095; %length of cable
9
10 %Show image and indicate endpoints and rider
11 imagesc([0,5.5098],[0,3.0993],flipud(I));set(gca,'YDir','normal');
12 axis equal; ylim([0,3.0993]); xlim([0,5.5098]); box on; hold on;
13 xlabel('$x$ (meters)','interpreter','latex'); ylabel('$y$ (meters)','interpreter','latex');
14 plot(L(1),L(2),'o','MarkerEdgeColor','b','MarkerFaceColor','b','MarkerSize',8);
15 plot(R(1),R(2),'o','MarkerEdgeColor','b','MarkerFaceColor','b','MarkerSize',8);
16 plot(M(1),M(2),'o','MarkerEdgeColor','b','MarkerFaceColor','b','MarkerSize',8);
17
18 %Define system of 6 equations to solve (c is vector containing 6 unknowns)
19 F1 = @(c) c(3) + c(2)*cosh((L(1)-c(1))/c(2)) - L(2); %Left cable must pass through left endpoint L
20 F2 = @(c) c(3) + c(2)*cosh((M(1)-c(1))/c(2)) - M(2); %Left cable must pass through rider at M
21 F3 = @(c) c(6) + c(5)*cosh((M(1)-c(4))/c(5)) - M(2); %Right cable must pass through rider at M
22 F4 = @(c) c(6) + c(5)*cosh((R(1)-c(4))/c(5)) - R(2); %Right cable must pass through right endpoint R
23 F5 = @(c) c(2)*(sinh((M(1)-c(1))/c(2))-sinh((L(1)-c(1))/c(2))) + c(5)*(sinh((R(1)-c(4))/c(5))-sinh((M(1)-
24 F6 = @(c) c(2)*cosh((M(1)-c(1))/c(2))- c(5)*cosh((M(1)-c(4))/c(5)); %Tension must be continuous at rider
25 cOut = fsolve(@(c) [F1(c);F2(c);F3(c);F4(c);F5(c);F6(c)],[-1,1,1,-1,1,1]);
26
27 %Plot theoretical shape of cable
28
29 %Left cable
30 xPlot = linspace(L(1),M(1),25);
31 yPlot = cOut(3) + cOut(2)*cosh((xPlot-cOut(1))/cOut(2));
32 plot(xPlot,yPlot,'r:','linewidth',4,'markersize',10);
33
34 %Right cable
35 xPlot = linspace(M(1),R(1),25);
```

```
35 xPlot = linspace(R(1),R(2),25);
36 yPlot = cOut(6) + cOut(5)*cosh((xPlot-cOut(4))/cOut(5));
37 plot(xPlot,yPlot,'b:','linewidth',4,'markersize',10);
38
39 %Straight lines for reference
40 plot([L(1),M(1)],[L(2),M(2)],'k','linewidth',2);
41 plot([M(1),R(1)],[M(2),R(2)],'k','linewidth',2);
```

▶ Run Script ?

Assessment: All Tests Passed

Submit ?

✔ cOut correct

Output

Equation solved (matlab: helpview('optim','eqn_solved','CSHelpWindow')).

fsolve completed because the vector of function values is near zero as measured by the value of the function tolerance (matlab: helpview('optim','fcn_tolerance_fsolve', the problem appears regular (matlab: helpview('optim','appears regular','CSHelpWindow')) as measure

4. Add a stationary rider

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